Roadmap. §54 (pp. 1–4): Maxwell theory from $\mathcal{L} = -\frac{1}{4}F^2 + J \cdot A$, gauge invariant only if $\partial_\mu J^\mu = 0$ ($J^\mu = e\bar{\Psi}\gamma^\mu\Psi$, the Noether current: LecNote22 pp. 6–11; pp. 6–10 skipped in class). §55 (pp. 4–9): Coulomb gauge, two transverse photons. §56 (pp. 9–10): LSZ, $Z_0(J)$, made covariant (Feynman gauge) in LecNote22 pp. 1–5; R_ξ is §62. Supplements: ClassicalFields, FieldTransformations, FeynmanPropagator, AsymptoticStates, PhotonGauges.

let 21.

Vector Field.

Recall E&M.

$$\vec{\nabla}\cdot\vec{E} = \rho$$
$$\vec{\nabla}\times\vec{B} - \dot{\vec{E}} = \vec{J}$$
$$\vec{\nabla}\times\vec{E} + \dot{\vec{B}} = 0$$
$$\nabla\cdot\vec{B} = 0$$

← can be solved by writing a scalar and vector potential.

$$\vec{E} = -\vec{\nabla}\varphi - \dot{\vec{A}}$$
$$\vec{B} = \vec{\nabla}\times\vec{A}$$

← clearly $\varphi \to \varphi + \dot{T}$, $\vec{A} \to \vec{A} - \vec{\nabla}T$ would make the solution unchanged and it is still a solution.

~~And there is no physical~~ ~~exist~~

In the relativistic version (E&M should have covered it). Vector field

$$A^\mu \equiv (\varphi, \vec{A}).$$

And the field strength

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu.$$
$$F^{\mu\nu} = -F^{\nu\mu}.$$

By comparison, we know that

$$F^{0i} = E^i$$

$$F^{ij} = \varepsilon^{ijk} B^k.$$

The first two of Maxwell's equations are then.

$$\partial_\nu F^{\mu\nu} = J^\mu.$$

where $J^\mu \equiv (\rho, \vec{J})$.

And $\underline{\partial_\mu \partial_\nu F^{\mu\nu}} = \partial_\mu J^\mu$

$= 0$

$\longrightarrow \dot{\rho} + \vec{\nabla}\cdot\vec{J} = 0$ the continuity equation.

no "-" sign here bc ∂ is the same metric

To make sure the lower two maxwell equation are satisfied, we need the following

Eqwps $\partial_\lambda F^{\mu\nu} = 0$ (And it is automatically satisfied)

Then the transformation that doesn't change the redundant resulting field now is $A'^\mu = A^\mu - \partial^\mu \Gamma$.

$$F'^{\mu\nu} = \partial^\mu A'^\nu - \partial^\nu A'^\mu = F^{\mu\nu} - (\partial^\mu \partial^\nu - \partial^\nu \partial^\mu)\Gamma.$$

as $\partial^\mu \partial^\nu = \partial^\nu \partial^\mu$

$F'^{\mu\nu} = F^{\mu\nu}$ and hence invariant under such gauge transformation.

$x^\mu = (t, \vec{x})$

$x_\mu = (-t, \vec{x})$

$\partial_\mu = \frac{\partial}{\partial x^\mu} = \frac{\partial}{\partial(t, \vec{x})}$ so no minus sign.

The following Lagrangian density leads to the Maxwell equations.

$$\mathcal{L} = -\frac{1}{4} F^{\mu\nu} F_{\mu\nu} + J^\mu A_\mu .$$

$-\frac{1}{4} F^{\mu\nu} F_{\mu\nu}$: Gauge Invariant

$J^\mu A_\mu$: ↓

$$J^\mu A'_\mu - J^\mu A_\mu$$

$$= \bar{J}^\mu \partial_\mu \Gamma$$

$$= (\partial_\mu \bar{J}^\mu)\Gamma - \partial_\mu (\bar{J}^\mu \Gamma)$$

$(\partial_\mu \bar{J}^\mu)\Gamma$: ↑ 0 because J^μ is a conserved current.

$\partial_\mu (\bar{J}^\mu \Gamma)$: ↑ total divergence which vanishes at infinite under normal/reasonable vanishing boundary conditions

So $\mathcal{L}$ is gauge transformation invariant.

$$\frac{\delta \mathcal{L}}{\delta A_\mu} = \frac{\partial \mathcal{L}(x)}{\partial A_\mu(x)} - \partial_\rho \frac{\partial \mathcal{L}(x)}{\partial(\partial_\rho A_\mu(x))}$$

$$= (g^{\mu\nu}\partial^2 - \partial^\mu \partial^\nu) A_\nu + J^\mu = 0$$

which is

$$\partial_\nu F^{\mu\nu} = +J^\mu$$ ← hence the Maxwell Equation

$$(g^{\mu\nu}\partial^2 - \partial^\mu\partial^\nu) A_\nu = g^{\mu\nu}\partial_\alpha\partial^\alpha A_\nu - \partial^\mu \partial^\nu A_\nu$$

$$= \partial_\alpha \partial^\alpha A^\mu - \partial^\mu \partial^\nu A_\nu = \partial_\alpha \partial^\alpha A^\mu - \partial^\mu \partial^\alpha A_\alpha = \partial_\alpha F^{\alpha\mu}$$

A_μ is identified as the photon field.

But here is one problem:
photon has two independent polarizations (as we measured the property of light) (it's essentially massless, gauge symmetry preserved (not spontaneously))

Current limits on it
$m_\gamma < 10^{-15}$ eV [1]
$= 1.8 \times 10^{-48}$ g

Let's see how we deal with it

$$\mathcal{L} = -\frac{1}{4} F^{\mu\nu} F_{\mu\nu} + J^\mu A_\mu$$

Since $F^{\mu\nu}$ is anti symmetric, there is no $\partial^0 A^0$ terms so no propagating (in time) for A^0 component and hence it is not dynamical
(no conjugate momentum for the field's field A^0)

1 Current (PDG) limit: $m_\gamma < 1 \times 10^{-18}$ eV $\approx 1.8 \times 10^{-51}$ g (solar-wind magnetic fields, Ryutov 2007), a thousand times below the 10^{-15} eV written here.

Also there is the redundancy of gauge transformation, that we need to eliminate.
(otherwise, imagine we do a path integral on all possible field configurations, there are infinite ways to capture the same physics!)
In principle, this doesn't matter so long as you known how to Module out the redundancy.

But we can also get rid of it by
"Gauge fixing". e.g.

Require $n^\mu A_\mu(x) = 0$
where n^μ is a Lorentz four vector.

If $n^2 > 0$, ~~time~~ space-like, it is axial gauge

$n^2 = 0$, light-cone gauge

$n^2 < 0$, time-like, temporal agauge.

Gauge Choices do not affect $\vec{E}, \vec{B}$ but remove dof. There is no need to uniquely define A^μ's value (shifting by const allowed)

$\partial_\mu A^\mu = 0$ ($k^\mu A_\mu(x) = 0$) Lorentz gauge.

and $\vec{\nabla} \cdot \vec{A}(x) = 0$, ~~Coloumb~~ Coulomb Gauge.
(also known as radiation gauge. or transverse gauge).
This corresponds to projection $A_i(x) \to (\delta_{ij} - \frac{\nabla_i \nabla_j}{\nabla^2}) A_i(x)$

1 Should read $A^\mu = (\varphi, \mathbf{A})$; $J^\mu = (\rho, \mathbf{J})$ (p. 2). The $-\rho\varphi$ is $J^0 A_0$ with $A_0 = -\varphi$.

In this gauge, we can further the calculation

$$\mathcal{L} = \frac{1}{2}\dot{\vec{A}}\cdot\dot{\vec{A}} - \frac{1}{2}\nabla_j A_i \nabla_j A_i + J_i A_i$$

$$+ \frac{1}{2}\nabla_i A_j \nabla_j A_i + \dot{A}_i \nabla_i \varphi.$$

$$+ \frac{1}{2}\nabla_i\varphi\nabla_i\varphi - \rho\varphi$$

IBP

$\nabla_i A_i = 0$

by the gauge choice

φ is $J^\mu = (\varphi, \vec{A})$ ①

vary φ, we get

$$-\nabla^2\varphi = \rho \rightarrow \varphi(x,t) = \int d^3y \frac{\rho(\vec{y},t)}{4\pi|\vec{x}-\vec{y}|}$$

this is unique solution for ρ & φ vanishing at infinity

so here φ has no dynamics, only given by the charge density.

$$\frac{1}{2}\nabla_i\varphi\nabla_i\varphi \rightarrow -\frac{1}{2}\nabla_i\varphi\nabla^2\varphi \rightarrow +\frac{1}{2}\varphi\rho$$

$$+(-\rho\varphi) = -\frac{1}{2}\rho\varphi.$$

$$\rightarrow \quad \mathcal{L}_{coul} = -\frac{1}{2}\int d^3y \frac{\rho(\vec{x},t)\rho(\vec{y},t)}{4\pi|x-y|}.$$

$\mathcal{T}$

Coulomb potential.

Note that this can only be done in Coulomb gauge. In other gauges, φ couples to $\nabla\vec{A}$, which would be none zero.

(In this gauge, for free field when $J_i = 0$.)

We can vary the Λ to find $\vec{E}$ and $\vec{B}$.

$$-\partial^2 A_i(x) = \left(\delta_{ij} - \frac{\nabla_i \nabla_j}{\nabla^2}\right) J_j(x) \quad (1)$$

one can insert the projection to ensure the gauge choice

1 The projector is not optional: Ampère's law in Coulomb gauge reads $-\partial^2 \mathbf{A} = \mathbf{J} - \nabla\dot{\varphi}$, and $-\nabla^2\varphi = \rho$ with $\dot{\rho} = -\nabla\cdot\mathbf{J}$ gives $\nabla\dot{\varphi} = \nabla\nabla^{-2}(\nabla\cdot\mathbf{J})$. So only the transverse current sources $\mathbf{A}$; the longitudinal part of $\mathbf{J}$ is carried by the instantaneous φ of p. 6.

→ we have A_i obeys the massless Klein Gordon Equation (not surprising!!)

And the general solution

(Recall scalar and spinor case)

$$\vec{A}(x) = \sum_{\lambda=\pm} \int \widetilde{dk} \left[\varepsilon^*_\lambda(\vec{k}) a_\lambda(\vec{k}) e^{-ikx} + \varepsilon_\lambda(\vec{k}) a^\dagger_\lambda(\vec{k}) e^{ikx} \right]$$

$\varepsilon_\pm(\vec{k})$ are the polarization vectors

With Columb gauge, we can identify them as two circular polarizations. ε_+ right-handed

ε_- left-handed.

with wave vector $\vec{k} = (0, 0, k)$, chosen along z-direction

One can set $\varepsilon_+(\vec{k}) = \frac{1}{\sqrt{2}}(1, -i, 0)$

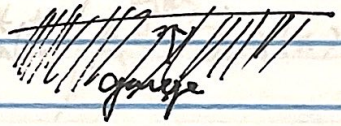
$\varepsilon_-(\vec{k}) = \frac{1}{\sqrt{2}}(1, +i, 0)$

but also more useful are the completeness relations, such that

$\vec{k} \cdot \varepsilon_\lambda(\vec{k}) = 0$

$\varepsilon_\alpha(\vec{k}) \cdot \varepsilon_\lambda^*(\vec{k}) = \delta_{\lambda\lambda'}$

$\sum_{\alpha=\pm} \varepsilon_{i\lambda}^*(\vec{k}) \varepsilon_{j\lambda}(\vec{k}) = \delta_{ij} - \frac{k_i k_j}{k^2}$ | Gauge dependent.



Just as before, one can Fourier Transform and project out the photon field's creation & annihilation operators. using the polarization vectors

$$a_\lambda(\vec{k}') = +i \varepsilon_\lambda(\vec{k}) \cdot \int d^3x \, e^{-ikx} \overleftrightarrow{\partial_0} \vec{A}(x)$$

$$a_\lambda^\dagger(\vec{k}') = -i \varepsilon_\lambda^*(\vec{k}) \cdot \int d^3x \, e^{ikx} \overleftrightarrow{\partial_0} \vec{A}(x).$$

here $f \overleftrightarrow{\partial_\mu} g = f(\partial_\mu g) - (\partial_\mu f) g$.

We can also see that (1)

1 Where these come from (§55): from p. 6's $\mathcal{L}$, $\Pi_i = \dot{A}_i$, but $[A_i, \Pi_j] = i\delta_{ij}\delta^3$ would contradict $\nabla \cdot \mathbf{A} = 0$. The consistent rule is $[A_i(\mathbf{x}, t), \dot{A}_j(\mathbf{y}, t)] = i\delta_{ij}^\perp(\mathbf{x} - \mathbf{y})$, the transverse δ: the Fourier transform of $\delta_{ij} - k_i k_j / \mathbf{k}^2$. With the projections above it is equivalent to p. 9's $[a, a^\dagger]$.

$$[a_\lambda(\vec{k}), a_{\lambda'}(\vec{k}')] = 0$$

$$[a^\dagger_\lambda(\vec{k}), a^\dagger_{\lambda'}(\vec{k}')] = 0$$

$$[a_\lambda(\vec{k}), a^\dagger_{\lambda'}(\vec{k}')] = (2\pi)^3 2\omega\, \delta^3(\vec{k}-\vec{k}')\, \delta_{\lambda\lambda'}$$

LSZ reduction

Good old trick

$$a_\lambda(+\infty) - a_\lambda(-\infty) = \int_{-\infty}^{+\infty} dt\, \partial_0 a_\lambda(\vec{k})$$

$$\to \varepsilon^*_\lambda(\vec{k}) \cdot \int A\, e^{-ikx} \overleftrightarrow{\partial_0} A(x)$$

...

$$= i\varepsilon^\mu_\lambda(\vec{k}) \int d^4x\, e^{-ikx} (-\partial^2) A_\mu(x) \quad (1)$$

here we promote $\vec{A}(x) \to A^\mu(x)$

by adding $\varepsilon^0_\lambda(\vec{k}) \equiv 0$. ~~for this gauge.~~ more later

Then we have the usual

$$\langle 0 | A^+(x) | 0 \rangle = 0$$

$$\langle k, \lambda | A^-(x) | 0 \rangle = \varepsilon^+_\lambda(\vec{k})\, e^{-ikx}$$

$$\langle k, \lambda | k', \lambda' \rangle = (2\pi)^3 2\omega\, \delta^3(\vec{k}-\vec{k}')\, \delta_{\lambda\lambda'}$$

This implies that we need to renormalize the theory.

$$\mathcal{L} = -\frac{1}{4} Z_3 F^{\mu\nu} F_{\mu\nu} + Z_1 J^\mu A_\mu. \quad (2)$$

1 The "...": $\partial_0[e^{-ikx} \overleftrightarrow{\partial_0} \mathbf{A}] = e^{-ikx}(\partial_0^2 + \omega^2)\mathbf{A}$, and $\omega^2 = \mathbf{k}^2 \to -\nabla^2$ by parts, so $\partial_0 a_\lambda = i\varepsilon_\lambda \cdot \int d^3x\, e^{-ikx}(-\partial^2)\mathbf{A}$ (with ε_λ, as on p. 8). Likewise $a^\dagger_\lambda(-\infty) - a^\dagger_\lambda(+\infty) = i\varepsilon^{\mu *}_\lambda \int d^4x\, e^{ikx}(-\partial^2)A_\mu$: incoming photons carry ε^*, outgoing ε (LecNote23 p. 1).

2 Why Z_3 and Z_1: interactions change the residue of the photon pole (LecNote23 p. 6), and Z_3 restores $\langle k, \lambda | A^i | 0 \rangle = \varepsilon^i e^{-ikx}$. $\langle 0 | A^i | 0 \rangle = 0$ is automatic (rotation invariance), and $\frac{1}{2}m^2 A_\mu A^\mu$ is not gauge invariant: no photon-mass counterterm. Z_1 goes with the coupling: for $J^\mu = e\bar{\Psi}\gamma^\mu\Psi$ (LecNote22 p. 10) it is the vertex counterterm (LecNote23 p. 4).

(10)

1 Covariance returns: with $n^\mu = (1,0,0,0)$, $\sum_\lambda \varepsilon_\lambda^{\mu *} \varepsilon_\lambda^\nu = g^{\mu\nu} + n^\mu n^\nu k^2/\mathbf{k}^2 + (\text{terms} \propto k^\mu \text{ or } k^\nu)$. Over $k^2 - i\epsilon$ the nn piece is $+n^\mu n^\nu/\mathbf{k}^2$ and cancels the instantaneous Coulomb term; the k^μ terms vanish against conserved currents, $k_\mu \tilde{J}^\mu = 0$. So $\tilde{\Delta}^{\mu\nu} \to g^{\mu\nu}/(k^2 - i\epsilon)$: Feynman gauge (LecNote22 pp. 1–5).

We can continue and evaluate the propagator

$$\frac{1}{i}\Delta^{ij}(x-y) \equiv \langle 0|T A^i(x) A^j(y)|0\rangle$$

$$= \frac{1}{i}\int \frac{d^4k}{(2\pi)^4} \frac{e^{-ik(x-y)}}{k^2 - i\varepsilon} \sum_{\lambda=\pm} \varepsilon_\lambda^{i*}(\vec{k}) \varepsilon_\lambda^j(\vec{k})$$

We now have to evaluate the path integral

$$Z_0[J] \equiv \langle 0|0\rangle_J = \int \mathcal{D}A\, e^{i\int d^4x \left[-\frac{1}{4}F^{\mu\nu}F_{\mu\nu} + \underline{J^\mu A_\mu}\right]}.$$

↑ viewed as external sources.

again A^0 is not dynamical as we can get

$$S_{coul} = -\frac{1}{2}\int d^4x\, d^4y\, \delta(x^0 - y^0) \frac{J^0(x) J^0(y)}{4\pi |\vec{x} - \vec{y}|}$$

$$Z_0[J] = \exp\left[i S_{coul} + \frac{i}{2}\int d^4x\, d^4y\, J_i(x) \Delta(x-y) J_j(y)\right].$$

$$\Delta^{\mu\nu}(x-y) \equiv \int \frac{d^4k}{(2\pi)^4} e^{-ik(x-y)} \widetilde{\Delta}^{\mu\nu}(k)$$

$$\widetilde{\Delta}^{\mu\nu}(k) \equiv -\frac{1}{k^2}\delta^{\mu 0}\delta^{\nu 0} + \frac{1}{k^2 - i\varepsilon} \sum_{\lambda=\pm} \varepsilon_\lambda^{\mu *}(\vec{k}) \varepsilon_\lambda^\nu(\vec{k}). \quad (1)$$

$$\int_{-\infty}^{+\infty} \frac{dk^0}{(2\pi)} e^{-ik(x^0 - y^0)} = \delta(x^0 - y^0)$$

$$\int \frac{d^3k}{(2\pi)^3} \frac{e^{-i\vec{k}(\vec{x}-\vec{y})}}{\vec{k}^2} = \frac{1}{4\pi|x-y|}.$$